LETTERS TO THE EDITORS

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Remanent Magnetism of the Rajmahal Traps of North-Eastern India

Following a palæomagnetic survey of the Deccan Traps of India[1–3], we have extended our investigations of Indian rock formation to other geological ages. In particular we have made rock magnetic measurements on samples of lavas from the Rajmahal Traps of eastern Bihar. These lavas, which consist mainly of basaltic and pitchstone trap flows, overlie the Upper Gondwana shales and sandstones. They are most probably Jurassic in age, and according to Hobson[4] they represent the earliest part of the volcanic activity that afterwards, during the late Cretaceous and early Eocene periods, gave rise to the Deccan Traps.

During January 1957 we collected samples of the rocks from five localities spaced about 20 miles apart. Approximately ten samples were taken from each site, usually over a distance of several hundred yards. In two of the localities the material was found in the beds of streams, while in the remaining three the blocks were obtained from freshly exposed rock in quarries. The samples were analysed in Bombay and London.

The specimens taken from the three quarries (33 blocks in all) were found to be consistently magnetized along an axis directed towards the west of north with a steep upward dip. A total of 16 blocks from the two stream beds, on the other hand, were badly weathered and showed widely scattered directions of magnetization, and the results for these particular samples have been rejected.

The magnetic direction for the samples from the quarry sites are plotted in Fig. 1 and details of the measurements are given in Table 1. The steep upward dip (−64°) is consistent with the assumption that the Indian land mass occupied a position south of the equator at the time when the rocks acquired their magnetization, and if the mean direction of magnetization is taken to correspond to that of a geomagnetic axial dipole field at the time when the rocks were formed, we must suppose that the Rajmahal area had a southern latitude of 46° at that time. The corresponding ancient position of the north geographical pole, computed on the polar wandering hypothesis, lies in the southern Carribbean at lat. 13° N., long. 70° W., as shown by point IX in Fig. 2.

It is interesting to compare this result with those published previously[1–3,6] for the lavas of the Deccan Trap series. The north pole positions obtained from measurements made on rocks from eight different sites in the Deccan formation are also shown in Fig. 2. They fall into three categories :

(1) Sites V–VIII, which belong to the Lower Traps, give pole positions farthest to the south. The rocks are all reversely magnetized (that is, in south-easterly directions) with rather steep downward dips.

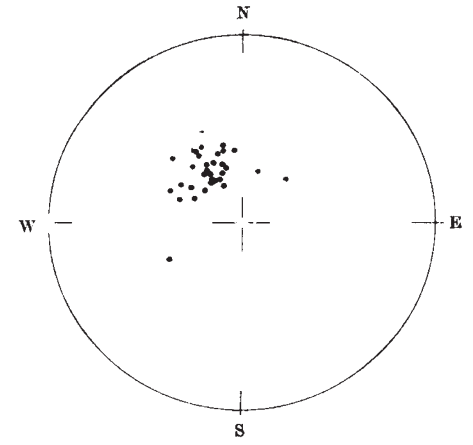


Fig. 1. Rajmahal area : ●, upward dips. Equal area projection

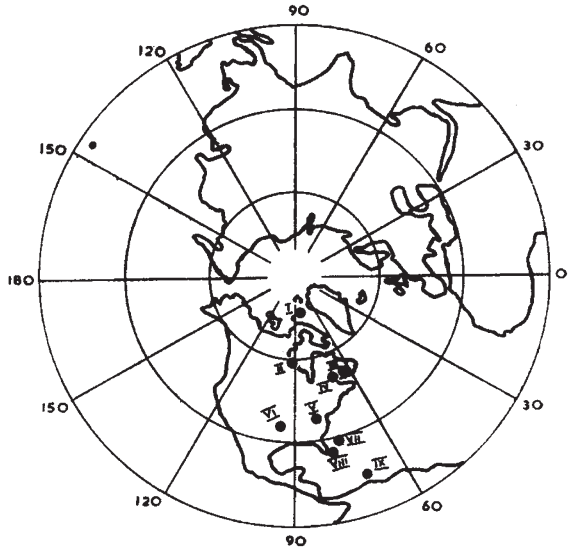


Fig. 2. Ancient pole positions for various sites : I, Pavagadh (upper) ; II, Pavagadh (lower) ; III, Amba (upper) ; IV, Nipani (upper) ; V, Linga ; VI, Nipani (lower) ; VII, Amba (lower) ; VIII, Khandala ; IX, Rajmahal

(2) Sites II–IV are all close to the tops of the Western Ghats, and give pole positions closer to the present geographical pole. The samples from these sites have normal magnetization (that is, towards the west of north), with considerably shallower upward dips.

(3) Site I is close to the top of Mount Pavagadh, and the rocks are acid tuffs believed to be more recent than the lavas of the main Trap sequence. Again the magnetic directions are normal, but the dips are almost zero.

It has been pointed out that this continuous decrease of dip with altitude above sea-level may be due to any one of three causes, namely :

(*a*) It may have occurred as a result of some mechanical effect associated with different depths of burial, such as that described by Graham[7].

Table 1. Rajmahal Rocks

Mean latitude of area	Mean longitude of area	No. of blocks	Mean intensity (c.g.s.u.)	Mean azimuth	Mean dip	Radius of 50 per cent circle of confidence*
25° 0′ N.	87° 51′ E.	33	$8{\cdot}85 \times 10^{-3}$	N. 328° E.	64° up	2°

* Measured by Fisher's method (ref. 5)

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(*b*) It may be indicative of partial magnetic instability which would cause a pulling of the magnetic directions towards that of the present Earth's field. This would tend to increase the magnetic dips of the reversely magnetized rocks (1) and to decrease those of the normally polarized ones (2) and (3).

(*c*) It may represent a continuous northward movement of India relative to the equator during the time when the Deccan Traps were being erupted.

The Rajmahal lavas discussed in this communication are older than the Deccan Traps. They have steeper magnetic dips, and give a pole position (IX) farther from the present pole than any of the Deccan rocks. Considering this steeper dip in relation to postulates (*a*), (*b*) and (*c*) above, it appears that: (*a*) it is unlikely to have been caused by compressional forces associated with deep burial, since all the specimens collected came from the uppermost 250 ft. of the Rajmahal series, and it is improbable that they have ever been overlain by any substantial thickness of material[4]; (*b*) it cannot have been increased by magnetic instability in the present Earth's field, as the Rajmahal rocks are normally magnetized, and this effect would tend to reduce rather than to increase their magnetic dips.

It seems, therefore, that the combined results for the Rajmahal and Deccan Traps conform best with the postulate (*c*) and are compatible with the supposition that there was a continuous northward drift of the Indian land mass extending from the Jurassic to the Eocene.

This conclusion must be regarded as tentative pending a more detailed survey of the areas concerned. The work is continuing, and a full account will be given elsewhere.

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The Triple Point of Argon as a Thermometric Fixed Point

In connexion with experimental work on the thermodynamic properties of binary mixtures of condensed gases, it was necessary to know accurately the triple point of argon on the absolute scale. Michels, Wassenaar, Sluyters and De Graaff[1] have recently made a careful determination of this temperature using a platinum resistance thermometer. They suggested that it should be used as a fixed point for thermometer calibrations, and since this proposal has much to recommend it, it is desirable that there should be an independent estimate of the triple point.

Using cylinder gases of the highest purity obtainable, further purified by fractional distillation, we have measured the vapour pressure of oxygen at the triple point of argon, the liquid oxygen being contained in a small bulb surrounded by argon which had just started to melt. The manometer used was that described by Lambert and Phillips[2], and individual measurements of the oxygen vapour pressure at the argon triple point were constant to $\pm$ 0·01 mm. The average value was 368·69 int. mm. of mercury with an estimated absolute uncertainty of $\pm$ 0·05 mm. Hoge[3] has given values of the vapour pressure of oxygen on a temperature scale based on a helium gas thermometer on which the oxygen boiling point is taken as 90·190° K. (with 0° C. $\equiv$ 273·160° K.). In the range involved, Hoge gives oxygen vapour pressures accurate to 0·1–0·2 mm., corresponding to temperatures accurate to 0·003–0·006°. From his data and the above figure for the oxygen vapour pressure, the triple point of argon is 83·819 $\pm$ 0·006° K. This is in good agreement with the value of Michels *et al.* of 83·819 $\pm$ 0·001° K.

Aston and Moessen[4] have recently carefully determined the normal oxygen boiling point and obtained a value of 90·154° K., which is 0·036° lower than the figure of the 1948 International Temperature Scale. With their value of the oxygen boiling point, our value for the argon triple point is 83·783 $\pm$ 0·006° K.

We have also made repeated measurements of the triple point pressure of argon, obtaining an average value of 516·82 $\pm$ 0·04 mm. The figure of 515·7 mm. given by Clark, Din and Robb[5] and used by them to obtain the triple point is undoubtedly too low.

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Raman Spectrum of Tetrakistrifluorophosphine-nickel

We have observed the Raman spectrum of liquid tetrakistrifluorophosphine-nickel, $Ni(PF_3)_4$, prepared as described by Wilkinson[1]. As required for the point group T_d, all the lines of the spectrum are found to be depolarized except three, which are highly polarized ($\rho \sim 0$). This observation provides strong evidence that the arrangement of the four attached phosphorus trifluoride molecules around the nickel atom is regular tetrahedral (cf. nickel tetracarbonyl).

The Raman frequencies of the three polarized lines are 954, 534 and 195 cm.$^{-1}$. They are clearly to be assigned respectively to ν_1 (mainly P–F stretching), ν_2 (mainly symmetrical 'umbrella' PF_3 deformation) and ν_3 (mainly Ni–P stretching). For comparison, the values of ν_1 and ν_2 for the phosphorus trifluoride molecule[2] are respectively 890 and 531 cm.$^{-1}$. Assuming a simple valency force field, the three polarized frequencies of $Ni(PF_3)_4$ may be used for deriving the two bond-stretching force constants k_{P-F} and k_{Ni-P} and the deformation force constant k_δ/l^2. The values thus obtained are $k_{P-F} = 7{\cdot}73$, $k_\delta/l^2 = 0{\cdot}53$ and $k_{Ni-P} = 2{\cdot}71$ (all in units of 10^5 dynes per cm.). The first two may be compared with the corresponding values for the phosphorus trifluoride molecule, $k_{P-F} = 5{\cdot}38$ and $k_\delta/l^2 = 1{\cdot}15$, as calculated from ν_1 and ν_2 assuming simple valency forces. The force constant of stretch-

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